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It Takes Two to Borrow: The Effects of the Equal Credit Opportunity Act on Housing, Credit, and Labor Market Decisions of Married Couples

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1 Big picture

- **Question.** What happens when antidiscrimination law relaxes income-related mortgage borrowing constraints by prohibiting lenders from discounting wives' incomes?
- **Policy shock.** The Equal Credit Opportunity Act (ECOA), signed in 1974 and effective in 1975, prohibited discrimination in lending based on sex and marital status.
- **Mechanism.** Before the reform, mortgage lenders commonly counted only half of the wife's income in joint applications. The reform raised the creditable share of the wife's income.
- **Empirical strategy.** Federal difference-in-differences uses the pre-reform wife income share as treatment intensity; state event studies exploit early state-level equal credit laws.
- **Main result.** The reform increased mortgage borrowing, homeownership, and house size for married households with working wives.
- **Labor supply channel.** The reform increased the return to wives' labor supply because additional earnings now raised mortgage borrowing capacity.
- **Model result.** A life-cycle model shows that allowing female labor supply to adjust approximately doubles the homeownership and borrowing effects of raising λ_d from 0.5 to 1.

2 Data and measurement

2.1 Historical setting and PSID sample

The ECOA banned sex and marital-status discrimination in credit. The paper uses PSID household panel data from 1969–1985, focusing on married households. The period begins when mortgage data become available and ends before the 1986 Tax Reform Act.

Interpretation. The treatment is a change in what income counts toward mortgage borrowing capacity, not a direct subsidy.

2.2 Outcomes and treatment intensity

Key outcomes are debt-to-income (DTI), debt-service-to-income (DSTI), housing-to-income (HTI), mortgagor status, homeownership, loan-to-value (LTV), house size, and wives' labor-force participation. The treatment intensity is the wife's average labor-income share in total household income over 1969–1971.

Interpretation. A higher pre-reform wife income share implies a larger relaxation of the DTI constraint when income discounting ends.

3 Models and empirical specifications

3.1 DTI constraint and income discounting

$$\phi(y^m, y^f) = \lambda_y(y^m + \lambda_d s^f y), \qquad y^f = s^f y.$$

Interpretation. Raising λ_d from 0.5 to 1 makes the wife's income fully creditable and relaxes the constraint more for households with larger wife income shares.

3.2 Federal difference-in-differences

$$Y_{ist} = \sum_{t=1969, t\neq 1971}^{1985} \beta_t \delta_t \, share_i^{pre} + \alpha_{st} + \gamma_i + \Gamma' X_{ist} + \varepsilon_{ist}.$$

Interpretation. The design compares high-wife-income-share and low-wife-income-share households within state-years and households over time.

3.3 State-level event study

$$Y_{ist} = \sum_j \beta_j D_{st}^j + \alpha_s + \delta_{rt} + \gamma_i + \Gamma' X_{ist} + \varepsilon_{ist}.$$

Interpretation. State-level timing provides an independent check on the federal design.

3.4 Life-cycle model borrowing constraint

$$\phi(h_j, y_j^m, y_j^f) = \min\{\lambda_h p h_j, \lambda_y (y_j^m + \lambda_d y_j^f)\}.$$

Interpretation. The model experiment changes λ_d from 0.5 to 1, matching the abolition of wife-income discounting.

4 Identification and empirical design

The federal design assumes that households with different pre-reform wife income shares would have followed parallel trends absent the ECOA. Household fixed effects absorb time-invariant household heterogeneity; state-year fixed effects absorb aggregate and state-specific shocks. State event studies use early state laws to validate the national results and show no strong pre-trends in key housing outcomes.

5 Tables and figures

5.1 Figure 1: Laws prohibiting discrimination in access to credit based on gender

Read: The figure places the ECOA in global perspective. In the early 1970s almost no countries had explicit legal protections against gender-based credit discrimination; the United States was an early mover.

Economic message. The ECOA is a historically important financial-inclusion reform.

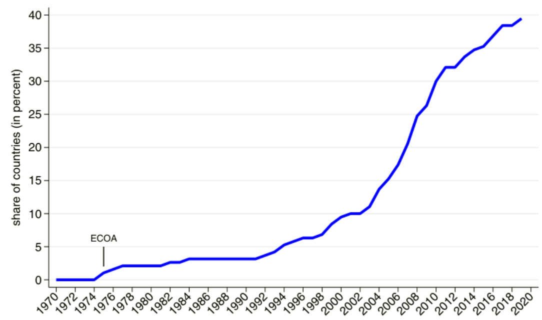

Figure 1
Laws prohibiting discrimination in access to credit based on gender
The graph shows the share of countries with explicit laws against discrimination in access to credit based on gender in a sample of 190 economies. The ECOA became effective in 1975 in the United States and Puerto Rico.
Source: World Bank Women, Business and the Law (WBL) database.

Figure 1: Laws prohibiting discrimination in access to credit based on gender

5.2 Figure 2: Housing and mortgage debt by wife's income contribution + Figure 3: Working wife share

Read: Figure 2 shows similar pre-reform trends across wife-income-share groups, then divergence after the reform. Figure 3 shows rising married female labor supply after the early 1970s.

Economic message. The descriptive evidence matches the DTI-channel hypothesis and motivates the labor-supply amplification mechanism.

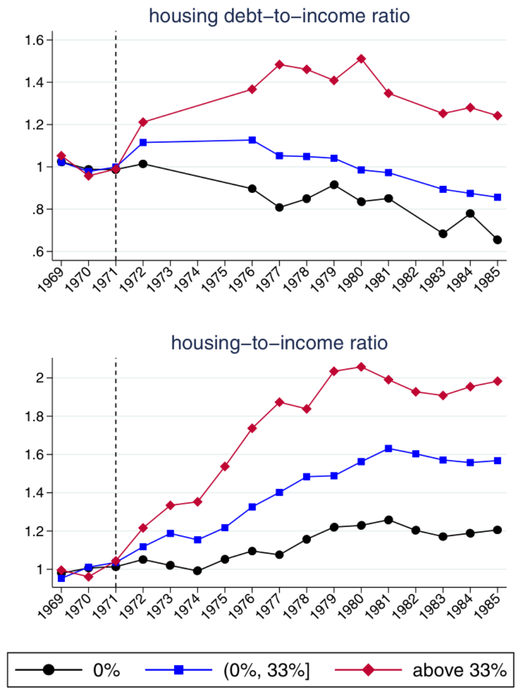

Figure 2
Housing and mortgage debt of married households by wife's income contribution
The left panel shows housing debt relative to income over time for married households with different average female income contributions in the pre-reform period (1969–1971). The right panel shows analogous results for housing relative to income. All series were normalized with their 1969–1971 average. The DTI and HTI ratios are winsorized at the 99th percentile within each year.

(a) Figure 2

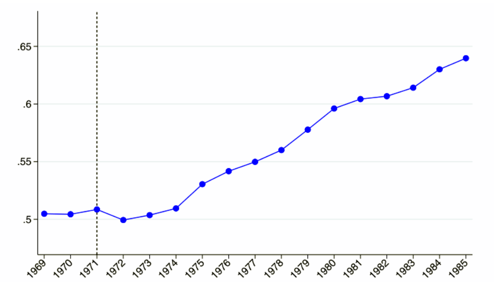

Figure 3
Share of married households with a working wife
The graph shows the share of married households in which the wife reports positive labor income over time. The data were smoothed by taking a 3-year moving average.

(b) Figure 3

5.3 Figure 4: Housing debt + Table 1: Housing debt

Read: Figure 4 shows positive post-reform effects on DTI, mortgagor status, and DSTI. Table 1 reports a positive DTI coefficient of 0.210*** and a positive mortgagor coefficient of 0.003***.

Economic message. The reform mainly moves households into mortgage borrowing rather than only raising balances of existing borrowers.

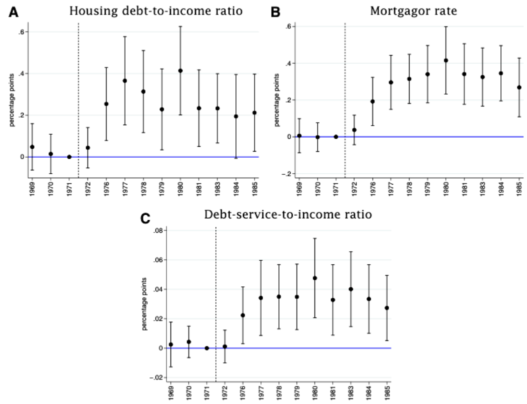

Figure 4
Housing debt
The graph presents the coefficients for the interaction term in Equation (1). The base year is 1971. The wife's income share and the DTI and DSTI ratios are defined as percentages. The DTI and DSTI ratios are winsorized at the 99th percentile within each year. Standard errors are clustered at the household-state level. The whiskers indicate 95% intervals.

(a) Figure 4: Housing debt

Table 1
Housing debt

	DTI	DTI, int. (log DTI)	DTI, ext. (mortgagor)	DSTI	DSTI, int. (log DSTI)
Post 1971	0.210***	−0.002	0.003***	0.025***	−0.001
× Tot. inc. share wife 71	(0.063)	(0.001)	(0.001)	(0.008)	(0.001)
Controls	yes	yes	yes	yes	yes
Household FE	yes	yes	yes	yes	yes
Time FE	yes	yes	yes	yes	yes
Mean	36.548	3.915	0.513	5.092	2.100
Observations	20,188	10,225	20,190	20,188	10,222

The table presents results for Equation (1), after replacing the year dummies with a dummy for the period after 1971. The abbreviations "int." and "ext." refer to the in- and extensive margin. Standard errors (in parentheses) are clustered at the household-state level ($^{*}p<.1$, $^{**}p<.05$, $^{***}p<.01$). The debt-to-income (DTI) and debt-service-to-income (DSTI) ratios are winsorized at the 99th percentile within each year. The wife's income share and the DTI and DSTI ratio are defined as percentages.

(b) Table 1: Housing debt

5.4 Figure 5: Housing + Table 2: Housing + Table 3: Binary interaction

Read: Figure 5 and Table 2 show positive effects on HTI, homeownership, and rooms. Table 3 reports a 0.083*** mortgagor-rate increase and a 0.062*** homeownership-rate increase for working-wife households after 1971.

Economic message. Credit access turns into real housing consumption and ownership.

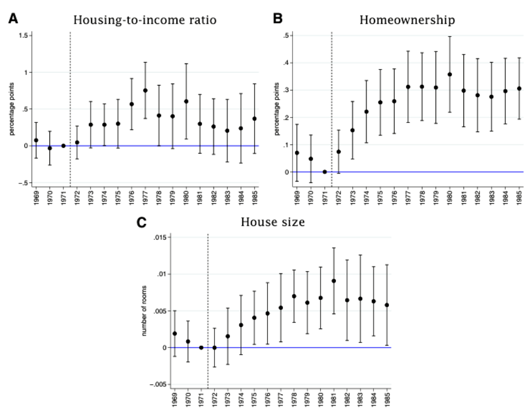

Figure 5
Housing
The graph presents the coefficients for the interaction term in Equation (1). The base year is 1971. The wife's income share and the HTI ratio are defined as percentages. The HTI ratio was winsorized at the 99th percentile within each year. Standard errors are clustered at the household-state level. The whiskers indicate 95% intervals.

(a) Figure 5: Housing

Table 2
Housing

	HTI	HTI, int. (log HTI)	HTI, ext. (homeowner)	LTV	Number of rooms
Post 1971	0.327***	0.001	0.002***	−0.117***	0.004**
× Tot. inc. share wife 71	(0.120)	(0.001)	(0.000)	(0.039)	(0.001)
Controls	yes	yes	yes	yes	yes
Household FE	yes	yes	yes	yes	yes
Time FE	yes	yes	yes	yes	yes
Mean	157.611	5.073	0.816	25.803	5.868
Observations	26,767	21,768	26,776	16,353	26,289

The table presents results for Equation (1), after replacing the individual year dummies with a dummy for the period after 1971. The abbreviations "int." and "ext." refer to the in- and extensive margin. Standard errors (in parentheses) are clustered at the household-state level ($^{*}p<.1$, $^{**}p<.05$, $^{***}p<.01$). The housing-to-income (HTI) and loan-to-value (LTV) ratios are winsorized at the 99th percentile within each year. The wife's income share and the HTI and LTV ratio are defined as percentages.

(b) Table 2: Housing

Table 3
Point estimates with binary interaction

	Mortgagor rate	Homeowner-ship rate
Dummy ×	0.083***	0.062***
post71 = 1	(0.021)	(0.014)
Controls	yes	yes
Household FE	yes	yes
Time FE	yes	yes
Mean	0.513	0.816
Observations	20,190	26,776

The table presents the results of estimating a binary version of Equation (1), where the interaction term is replaced by the interaction between a dummy for whether the average pre-reform share of the wife's labor income was positive and a dummy for the post-reform period after 1971. Standard errors are given in parentheses and are clustered at the household and state level (* p<.1, ** p<.05, *** p<.01).

Table 3: Point estimates with binary interaction

5.5 Figure 6: Event studies housing + Figure 7: Wives' probability to work

Read: Figure 6 uses state-level timing and shows post-reform increases in HTI, homeownership, and house size. Figure 7 shows suggestive labor-supply increases, especially for young renters with high propensity to respond.

Economic message. State-level evidence supports the housing mechanism and suggests that the reform raised the return to wives' labor supply.

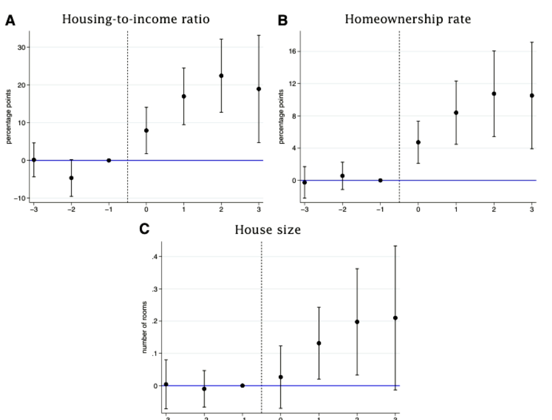

Figure 6
Event studies housing
The graph shows the coefficients for the treatment indicators D_{st}^{j} from Equation (2). The sample was restricted to households in which the wife had a positive average labor income over the 3 years prior to the event. Observations with $t \geq e_s + H$ are excluded, where H is the gap between the latest and earliest event year. Standard errors are clustered at the household-state level. The whiskers indicate 95% intervals.

(a) Figure 6

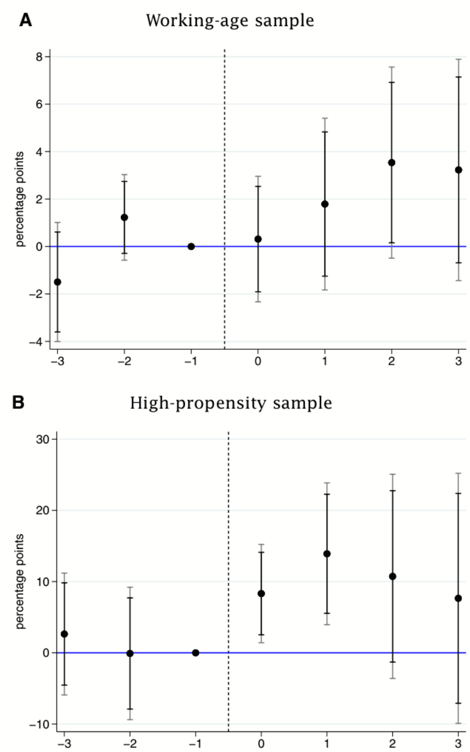

Figure 7
Wives' probability to work
The graph shows the coefficients for the treatment indicators D_{st}^{j} from Equation (2). The outcome is a dummy for whether the wife works more than 500 hours per year. The left panel shows results for all couples with wives below age 55 ("working-age" sample). In the right panel, the sample is restricted to couples with a wife below 35 who were renting prior to the reform, and the wife was working at most annual 500 hours ("high-propensity" sample). Standard errors are clustered at the household-state level. The gray (black) whiskers indicate 95% (90%) intervals.

(b) Figure 7

5.6 Table 4: Calibrated parameters + Table 5: Targeted and untargeted moments

Read: Table 4 sets $\lambda_h = 1$, $\lambda_y = 2$, and $\lambda_d = 0.5$ before the reform. Table 5 shows that the model matches homeownership, mortgagor, and FLFP rates across age groups and reasonable untargeted moments.

Economic message. The model is disciplined by the moments most relevant for the policy experiment.

Table 4
Calibrated parameters

Name	Value	Definition	Target/source
A. Externally calibrated parameters			
λ^h	1	LTV limit	PSID
λ^y	2	DTI limit	PSID
λ^d	0.5	Income discounting factor	ECOA
q	0.53	Net rental cost	PSID
$p_1 h_j$	11	House value $h_j = 1$	PSID
$p_2 h_j$	13.5	House value $h_j = 2$	PSID
r^s	0.043	Interest rate on savings	Federal Reserve
r^b	0.063	Interest rate on debt	SCF+
σ	2	CRRA parameter	
β	0.94	Discount factor	
b	0.7	Pension replacement rate	Attanasio et al. (2012)
B. Internally calibrated parameters			
ψ	1.63	Labor elasticity parameter	
μ^h	2.51e-3	Basic housing preference parameter	
ϕ^h	5.11e-3	House size preference parameter	
F_b	0.04	Buying cost	
F_s	0.07	Selling cost	
$\theta^f(j)$	see Internet Appendix D.2	Female leisure preference profile	
$\chi(j)$	see Internet Appendix D.2	Housing utility profile	

The table summarizes the externally (panel A) and internally (panel B) calibrated parameters (see text for details).

(a) Table 4

Table 5
Targeted and untargeted moments

Moment	Data	Model
A. Targeted moments		
Homeownership, 25-34	0.58	0.59
Homeownership, 35-44	0.78	0.78
Homeownership, 45-54	0.84	0.84
Homeownership, 55-64	0.8	0.84
Mortgagor rate, 25-34	0.55	0.53
Mortgagor rate, 35-44	0.68	0.7
Mortgagor rate, 45-54	0.57	0.66
Mortgagor rate, 55-64	0.35	0.31
FLFP, 25-34	0.43	0.45
FLFP, 35-44	0.39	0.36
FLFP, 45-54	0.42	0.4
FLFP, 55-64	0.39	0.38
B. Untargeted moments		
Average intensive-margin female hours	31	28
Labor supply elasticity	1.37-1.93	1.68
FLFP, homeowners	0.4	0.42
FLFP, mortgagors	0.41	0.39
Homeownership, above-median income	0.81	0.89
Homeownership, below-median income	0.66	0.65
Mortgagor rate, above-median income	0.67	0.69
Mortgagor rate, below-median income	0.42	0.42
FLFP, above-median income	0.51	0.51
FLFP, below-median income	0.32	0.32

Panel A compares the targeted moments (average homeownership, mortgagor, and FLFP rate by age group) in the model and the data. Panel B reports moments that were not explicitly targeted in the calibration. The FLFP rate in the data refers to women working at least 500 hours per year.

(b) Table 5

5.7 Figure 8: Comparison to data + Tables 6–7: Wage profiles and policy experiment

Read: Figure 8 shows that the simulated life-cycle profiles match the data reasonably well. Table 6 reports wage-process parameters. Table 7 shows that raising λ_d from 0.5 to 1 increases homeownership by 4.4 pp with fixed labor supply and 9.1 pp with flexible labor supply.

Economic message. Labor supply approximately doubles the homeownership and borrowing effects.

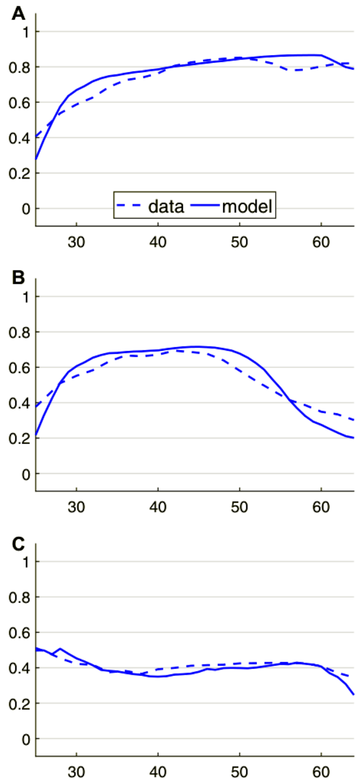

Figure 8
Comparison to data
The graph compares the homeownership rate, the share of mortgagors and the FLFP rate over the life cycle from the model to the corresponding life cycle profiles from the PSID for the period 1969-1971. The FLFP rate in the data refers to women working at least 500 hours per year. The data profiles were smoothed by taking a 3-year moving average.

Figure 8: Comparison to data

Table 6
Parameters of wage profiles

Names	Values
$\alpha_0^m, \alpha_1^m, \alpha_2^m, \alpha_3^m, \alpha_4^m$	−1.5381,0.4036,−0.0132,0.0002,−1.1439e-6
$\alpha_0^f, \alpha_1^f, \alpha_2^f, \alpha_3^f, \alpha_3^f$	−0.7490,0.2889,−0.0085,0.0001,−4.8191e-7
$\sigma^2_{\varepsilon^m}, \sigma^2_{\varepsilon^f}$	0.0055, 0.0064
$\mathcal{P}^{50}, \mathcal{P}^{75}$	0.8301, 0.9368

The table shows the estimated coefficients of the logarithmic wage processes from Equation (3) and the estimated part-time penalties $\mathcal{P}^{50}$ and $\mathcal{P}^{75}$. See the main text for details.

(a) Table 6

Table 7
Comparison: $\lambda^d = 0.5$ **versus** $\lambda^d = 1$

Variable	Effect with fixed labor supply	Effect with flexible labor supply
homeownership	0.044	0.091
mortgagor rate	0.043	0.087
FLFP	0	0.02
hours (intensive margin)	0	−0.378

The table shows the average changes in homeownership, the share of mortgagors and female labor force participation (in percentage points) as well as the change in average intensive margin female hours worked among households aged 25 to 64 when increasing the income discounting factor λ^d. from 0.5 to 1. The first column shows the respective differences when holding female labor supply choices constant at the optimal values from the no-reform scenario with $\lambda^d = 0.5$, whereas the second column shows the results when allowing female labor supply to adapt to its optimal values under the post-reform scenario.

(b) Table 7

5.8 Figure 9: Model experiments versus natural experiment + Figure 10: Life-cycle effects of raising λ_d

Read: Figure 9 compares model-implied effects to empirical estimates. Figure 10 shows that the reform effects are strongest for ages 25–34, the typical first-home-buying ages.

Economic message. The reform changed the timing of homeownership and labor supply over the life cycle.

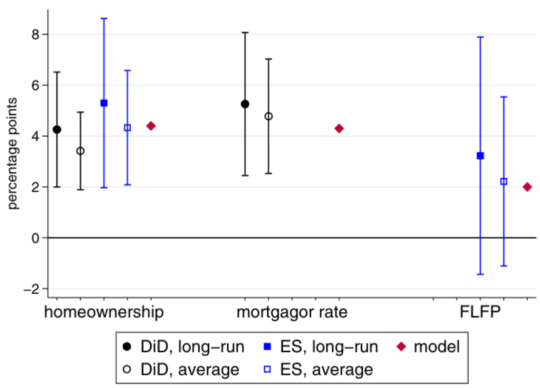

Figure 9
Comparison of model experiments with natural experiment in the data
The graph compares the model results to those from the data. For the homeownership and borrowing rate, the results from the "restricted" experiment in column 1 of Table 7 are compared to the results of the binary difference-in-differences (DiD) model (Figure B.2 in the Internet Appendix and the event study (ES) model (Figure 6, panel B), scaled by the share of households in the treatment group in 1971 (around 50%). For female labor supply, the result from the "unrestricted" model experiment in column 2 of Table 7 is compared to the event-study results in Figure 7, panel A. "Long-run" refers to the last point estimate in the respective DiD or ES graphs, and "average" refers to the average across all post-reform periods.

(a) Figure 9

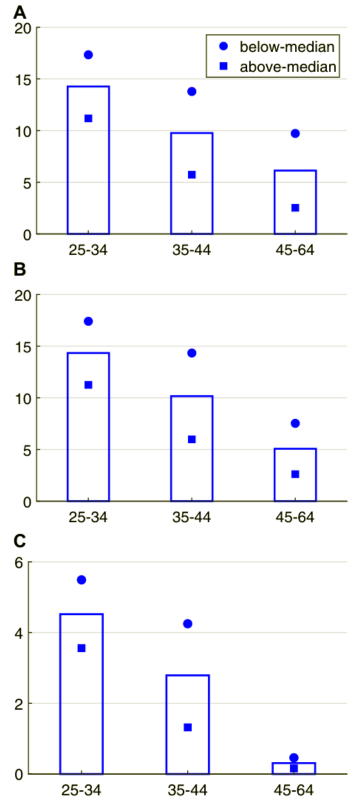

Figure 10
Life cycle effects of raising λ^d from 0.5 to 1
The graph shows the change in the homeownership rate, the share of mortgagors and the FLFP rate in percentage points over the life cycle when increasing the female income discounting factor λ_d from 0.5 to 1. The dots (squares) represent the respective changes in percentage points for households with below-median (above-median) male life cycle income.

(b) Figure 10

6 Short summary

- The ECOA relaxed mortgage DTI constraints by prohibiting lenders from discounting wives' incomes.
- Households with larger wife income shares benefited more.
- The reform increased mortgage borrowing, homeownership, and house size.
- State event studies support the federal evidence.
- Wives' labor-force participation rose, especially among young high-propensity households.
- The life-cycle model shows that flexible female labor supply roughly doubles the housing and borrowing effects.

7 Expanded conclusion

7.1 Core conclusion

The ECOA had large effects on married couples' mortgage access and homeownership by ending the discounting of wives' incomes in joint applications.

7.2 Economic mechanism

The mechanism is a DTI constraint. Before the reform, wives' incomes entered borrowing capacity with a low weight. After the reform, the income became fully creditable, especially benefiting households in which the wife earned a sizable share of income.

7.3 Labor-supply amplification

The reform raised the value of wives' market work because earnings now increased mortgage eligibility. The model shows that this channel roughly doubles the effects on homeownership and borrowing.

7.4 Policy interpretation

Gender-based credit restrictions delayed wealth accumulation through homeownership. Removing them improved access to credit and changed household labor-market incentives.

7.5 Limitations and future work

The labor-supply evidence is less precise than the housing evidence, and the model abstracts from some institutional details. Future work can study within-household bargaining, home-equity insurance, and modern lending technologies.